Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/12/2016
PI Name:	Grabham, Peter Ph.D.		
Project Title:	Effects of Space Radiation on the Function of Human Brain Capillaries and the Blood-brain-barrier: Implications for Neurodegenerative Disorders		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10032-3702	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
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No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2015 for reporting purposes (Ed., 7/14/15) NOTE: End date changed to 9/30/2015 per NSSC information (Ed., 8/29/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ponnaiya, Brian (Columbia University) Smilenov, Lubomir (Columbia University)		
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Task Description:	This proposal is aimed at determining the effects of space radiation on the function of human brain blood vessels (capillaries) and the Blood-Brain-Barrier (BBB). The brain vascular system is crucial to healthy functioning of the brain and its dysfunction is not only a primary event in a range of neurodegenerative diseases but also an important influencing factor in many others. Furthermore, vessel pathologies involve common susceptibility factors such as oxidative stress and inflammation so any radiation effects on capillaries can be influenced by other preexisting conditions. Here, we propose to investigate the effects of space radiation on angiogenesis and BBB function using human brain blood vessel cells in 2-dimensional and 3-dimensional tissue models. Since the brain vascular system is essentially an integrated multi-cell system we also propose to utilize other human brain cell types in tissue models to create the most realistic human vessel model possible. We have focused on the main aspects of vessel function that are relevant to the research emphases of acute and late risks to the brain. That is, functioning of the BBB, blood vessel development, and the compound effects of preexisting conditions such as inflammation. A correctly functioning blood system supplies the brain and other tissues with oxygen and nutrients while preventing deleterious blood-borne molecules from entering the brain and causing possible damage to brain cells. Tight junctions between the blood vessel cells are formed by proteins that make a tight seal and prevent leakiness. We will determine the effect of space radiation on blood vessels of the brain and its possible impact on neurodegeneration by testing for vessel formation (angiogenesis), vessel leakiness, and the influence of other factors such as inflammation molecules after exposure to both galactic cosmic radiation and radiation from the sun.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gamma radiation studies Studies to optimize assay conditions using gamma radiation revealed a previously unknown effect of gamma photons on human blood vessels. It is the first demonstration of a transient and rapid effect of gamma radiation on human endothelial barriers that involves cell detachment and loss of the cell adhesion molecule - PECAM-1. This effect occurred at the fractionated radiotherapy dose of 2 Gy and therefore has the potential to be of clinically relevant importance considering the role of PECAM-1 in many of the cellular processes involved in pathologies seen at longer times after radiation therapy. As such it could be a target for treatments that alleviate these pathologies. A second potential benefit is that of drug delivery to tumors. A temporary opening of the endothelial barrier by the dual effects of low temperatures and gamma radiation could be a means by which impermeable chemotherapy drugs could be delivered directly to tumors. Tissue models The development of 3-Dimensional human tissue models from normal human cells and stem cells has great potential in many fields of medical research. Tissue models can more accurately depict human tissue since the cells can be arranged spatially as they would be in vivo and can interact with each other as they would in the human body. A neurovascular unit can be used for basic research on many aspects of the human brain. These include regeneration, synaptic function, and degeneration. Because the tissue model is derived from individual cells, each cell type can be altered genetically before it is incorporated into the model.
	FINAL REPORTING July 2016: This is the final report of this study to determine the potential hazard of space radiation to human blood capillaries or microvessels. The research has been successful in identifying which species and doses of charged particle radiation are harmful to human blood capillaries and therefore constitute a risk of degenerative diseases in the space environment. The two critical functions of microvessels – Angiogenesis and Endothelial Barrier Function were examined. Photon radiation was used as a reference radiation and to perfect the assays before using beam time at NASA Space Radiation Laboratory (NSRL). A variety of low LET (linear energy transfer) and high LET charged particles were tested for their effect. Both functions were affected by space radiation (detailed below) with a higher relative biological effect than photons. These studies are the first to establish the deleterious effects of space radiation on human capillaries and form the foundation for further mechanistic and translational-based research relating to neurodegeneration. The work carried out in this grant award has led to several findings, publications, and further grant awards. Several papers were published and three more in preparation will eventually be submitted for publication. As such the grant award has been productive and successful. For angiogenesis we have completed studies that determine the dose limits for human vessel growth. It was found that high linear energy transfer (LET) Fe particles inhibit the growth of human vessels 50% with a dose of 40 cGy. However, Low LET protons also inhibited vessel growth 50% with a dose of 40 cGy. The unexpected high relative biological effectiveness (RBE) for protons compared to Fe ions is explained by our finding that each type of space radiation inhibits angiogenesis.by distinct mechanisms. This work was published in the vascular biology journal.–Vascular Cell (Grabham et al., 2013). A minireview entitled-The Effects of Radiation on Angiogenesis (Grabham and Sharma 2013) was also published in Vascular Cell. Follow up studies using different ion species to determine the LET ranges for these 2 types of inhibition have been completed. Low LET charged particles inhibit the early stages of vasculogenesis when tip cells have motile protrusive structures and are creating pioneer guidance tunnels through the matrix. High LET charged particles do not affect the early stages of vasculogenesis but they do affect the later stages when the endothelial cells migrate to form tubes. The addition of phorbol ester was shown to rescue the effect of low LET particles but not of heavy particles. This highlights the distinct nature of each mechanism. Heavy ions with a LET of 8 or greater appear act by the same mechanism. Ions with an LET of 2.9 or lower also act by the same mechanism albeit a different one than heavy ions. The same LET ranges were found in examining the activity of motile tips. Low LET particles inhibited motile tips whereas heavy ions did not. For Endothelial barrier function, preliminary experiments to test our assay system using gamma rays revealed a novel effect. Radiation caused the acute uncoupling of the cell adhesion molecule –PECAM-1 and the production of endothelial microparticles, an effect that is potentially clinically important in gamma radiation therapy. This work was published in the vascular biology journal–Microvascular Research (Sharma et al., 2013). A follow up study to determine if other low LET radiations cause the same response is now completed. Photons but not low LET charged particles induce the acute uncoupling of PECAM-1 and compromised barrier function. During the last year this paper was reworked to include more data and will be submitted shortly. Studies on high LET Fe ions are also completed. Fe ions were found to compromise barrier function and disrupt tight junctions with the most sensitive effect seen at 25 cGy. A

	manuscript describing this work is now published (Sharma et al., 2014). Further experiments on the later effect of protons on barrier function are now completed and demonstrate that barrier dysfunction is disrupted by low LET radiation at much higher doses than Fe particles, doses that coincide with the onset of apoptosis (see below). This work together with comparisons with the effects gamma rays is contained in a manuscript in preparation.
Task Progress:	In addition to completing studies on the effects of low LET charged particles and completing the LET range studies and angiogenesis, we have carried out additional studies to further the understanding of the effects of space radiation on barrier function and angiogenesis. Studies that begin to address mechanisms of charged particle effects on the microvasculature. These include, 1) The effects of another heavy ion species–Silicon ions on endothelial barrier function. 2) Oxidative stress pathways activated on 3D tissue models. 3) Rescue of barrier function by antioxidants. ANNUAL REPORTING July 2015: This grant became active in October 2011, and the data generated so far has contributed to fulfilling a number of the original aims directed at the 2 key functions of the vasculature: Angiogenesis and Endothelial Barrier Function. The harmful effects of Space Radiation on the structure and function of the human vascular tissue models have now been determined. For angiogenesis we have completed studies that determine the dose limits for human vessel growth. It was found that high linear energy transfer (LET) Fe particles inhibit the growth of human vessels 50% with a dose of 40 cGy. However, Low LET protons also inhibited vessel growth 50% with a dose of 40 cGy. The unexpected high relative biological effectiveness (RBE) for protons compared to Fe ions is explained by our finding that each type of space radiation inhibits angiogenesis.by distinct mechanisms. This work was published in the vascular biology journal.–Vascular Cell (Grabham et al., 2013). A minireview entitled-The Effects of Radiation on Angiogenesis (Grabham and Sharma 2013) was also published in Vascular Cell. Follow up studies using different ion species to determine the LET ranges for these 2 types of inhibition have been completed and a manuscript is in preparation. For Endothelial barrier function, preliminary experiments to test our assay system using gamma rays revealed a novel effect. Radiation caused the acute uncoupling of the cell adhesion molecule –PECAM-1 and the production of endothelial microparticles, an effect that is potentially clinically important in gamma radiation therapy. This work was published in the vascular biology journal–Microvascular Research (Sharma et al., 2013). A follow up study to determine if other low LET radiations cause the same response is now completed. Photons but not low LET charged particles induce the acute uncoupling of PECAM-1 and compromised barrier function. During the last year this paper was reworked to include more data and will be submitted shortly. Studies on high LET Fe ions are also completed. Fe ions were found to compromise barrier function and disrupt tight junctions with the most sensitive effect seen at 25 cGy. A manuscript describing this work is now published (Sharma et al., 2014). Further experiments on the later effect of protons on barrier function are now completed and demonstrate that barrier dysfunction is disrupted by low LET radiation at much higher doses than Fe particles, doses that coincide with the onset of apoptosis (see below). This work together with comparisons with the effects gamma rays is contained in a manuscript in preparation. By the end of this no-cost extension year and the grant as a whole there will be 7 publications that will collectively comprise of a completed body of work on the effects of ionizing radiation on angiogenesis and endothelial barrier function in human tissue models. In addition to completing studies on the effects of low LET charged particles and completing the LET range studies and angiogenesis, we have carried out additional studies to further the understanding of the effects of space radiation on barrier function and angiogenesis. Studies that begin to address mechanisms of charged particle effects on the microvasculature. These include, 1) The effects of another heavy ion species–Silicon ions on endothelial barrier function. 2) Oxidative stress pathways activated on 3D tissue models. 3) Rescue of barrier function by antioxidants.
Bibliography Type:	Description: (Last Updated: 03/04/2024)
Articles in Peer-reviewed Journals	Templin T, Sharma P, Guida P, Grabham P. "Short-term effects of low-LET radiation on the endothelial barrier: Uncoupling of PECAM-1 and the production of endothelial microparticles." Radiat Res. 2016 Dec;186(6):602-613. https://doi.org/10.1667/RR14510.1 ; PubMed PMID: 27905868 , Dec-2016
Articles in Peer-reviewed Journals	Wuu Y-R, Hu B, Okunola H, Paul AM, Blaber EA, Cheng-Campbell M, Beheshti A, Grabham P. "LET-dependent low dose and synergistic inhibition of human angiogenesis by charged particles: Validation of miRNAs that drive inhibition." Science. 2020 Dec 18;23(12):101771. https://doi.org/10.1016/j.isci.2020.101771 ; PMID: 33376971; PMCID: PMC7756138 , Dec-2020